

Problem Set 6

Kerri Riley

```
# Import modules  
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tibble' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

Warning: package 'purrr' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.2

-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --

v dplyr 1.2.1 v readr 2.2.0

v forcats 1.0.1 v stringr 1.6.0

v ggplot2 4.0.3 v tibble 3.3.1

v lubridate 1.9.5 v tidyr 1.3.2

v purrr 1.2.2

-- Conflicts ----- tidyverse_conflicts() --

x dplyr::filter() masks stats::filter()

x dplyr::lag() masks stats::lag()

i Use the conflicted package (<<http://conflicted.r-lib.org/>>) to force all conflicts to become

```
library(haven)

# Read the GSS data
gss <- read_dta("~/Documents/PAI741/Problem Set 6/Data/gss7224_r1.dta")
```

Recoding for Replication

```
# spkath, colath, libath
gss <- gss|>
  mutate(sp_kath = case_when(
    sp_kath == 1 ~ 0,
    sp_kath == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(colath = case_when(
    colath == 4 ~ 0,
    colath == 5 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(libath = case_when(
    libath == 1 ~ 0,
    libath == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

# Racists
# spkrac, colrac, librac
gss <- gss|>
  mutate(sp_krac = case_when(
    sp_krac == 1 ~ 0,
    sp_krac == 2 ~ 1,
    # The rest is missing
```

```

    TRUE ~ NA
  ))

gss <- gss|>
  mutate(colrac = case_when(
    colrac == 4 ~ 0,
    colrac == 5 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(librac = case_when(
    librac == 1 ~ 0,
    librac == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

# Communists
# spkcom, colcom, libcom
gss <- gss|>
  mutate(spkcom = case_when(
    spkcom == 1 ~ 0,
    spkcom == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(colcom = case_when(
    colcom == 4 ~ 0,
    colcom == 5 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(libcom = case_when(
    libcom == 1 ~ 0,
    libcom == 2 ~ 1,

```

```

    # The rest is missing
    TRUE ~ NA
  ))

# militarists
# spkmil, colmil, libmil
gss <- gss|>
  mutate(spkmil = case_when(
    spkmil == 1 ~ 0,
    spkmil == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(colmil = case_when(
    colmil == 4 ~ 0,
    colmil == 5 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(libmil = case_when(
    libmil == 1 ~ 0,
    libmil == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

# Homosexuals
# spkhomo, colhomo, libhomo
gss <- gss|>
  mutate(spkhomo = case_when(
    spkhomo == 1 ~ 0,
    spkhomo == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

```

```

gss <- gss|>
  mutate(colhomo = case_when(
    colhomo == 4 ~ 0,
    colhomo == 5 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(libhomo = case_when(
    libhomo == 1 ~ 0,
    libhomo == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

```

Replicate Figure 1

```

# Replicate figures
# Figure 1: 0-15 GSS Modified Stouffer Tolerance Battery, 1976-2024

# Create intolerance score by adding together the sum of all the values for the respondent.
gss <- gss |>
  mutate(intolerance_score = spkath + colath + libath +
    spkrac + colrac + librac +
    spkcom + colcom + libcom +
    spkmil + colmil + libmil +
    spkhomo + colhomo + libhomo)

# Take the mean score of all respondents by year
figure1 <- gss |>
  group_by(year) |>
  summarize(mean_intolerance = mean(intolerance_score, na.rm = TRUE))

# Take figure1 and filter
figure1 <- figure1 |>
  filter(!is.nan(mean_intolerance))

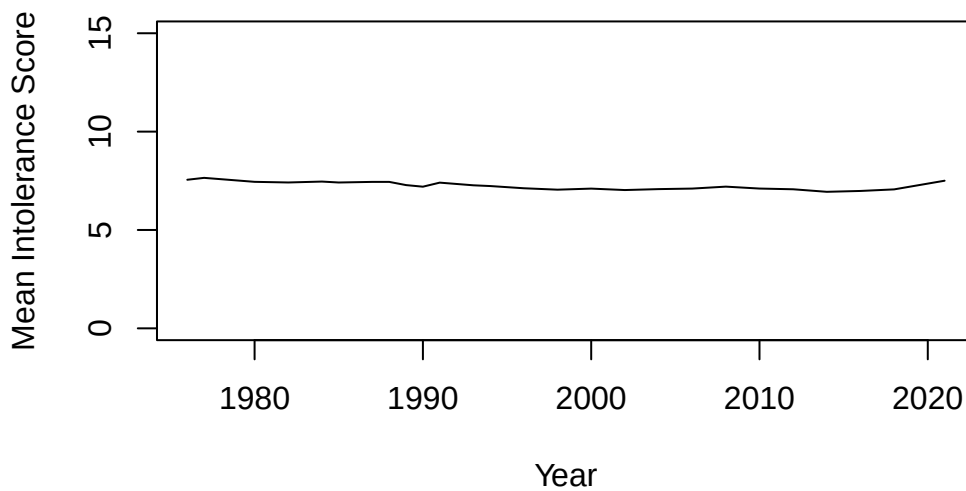
# Show table of data for figure1
figure1

```

```
# A tibble: 26 x 2
  year      mean_intolerance
  <dbl>+<lbl>      <dbl>
1 1976             7.55
2 1977             7.65
3 1980             7.45
4 1982             7.41
5 1984             7.46
6 1985             7.41
7 1987             7.44
8 1988             7.44
9 1989             7.28
10 1990            7.20
# i 16 more rows
```

```
# Create plot 1
plot(figure1$year, figure1$mean_intolerance,
     # Setting theplot type to a line plot
     type = "l",
     # Main title of figure
     main = "The 0-15 GSS Modified Stouffer Tolerance Battery, 1976-2024",
     # X-axis label
     xlab = "Year",
     # Y-axis label
     ylab = "Mean Intolerance Score",
     # Parameters for y-axis
     ylim = c(0,15)
)
```

The 0–15 GSS Modified Stouffer Tolerance Battery, 1976–20



Replicate Figure 2

```
# Figure 2: Intolerance Towards Racists and Homosexuals, 1976–2024

gss <- gss |>
  mutate(
    racist_score = spkrac + colrac + librac,
    homosexual_score = spkhomo + colhomo + libhomo
  )

figure2 <- gss |>
  group_by(year) |>
  summarize(
    racist_mean = mean(racist_score, na.rm = TRUE),
    homosexual_mean = mean(homosexual_score, na.rm = TRUE)
  )

# Remove years without data
figure2 <- figure2 |>
  filter(!is.nan(racist_mean) & !is.nan(homosexual_mean))
```

```

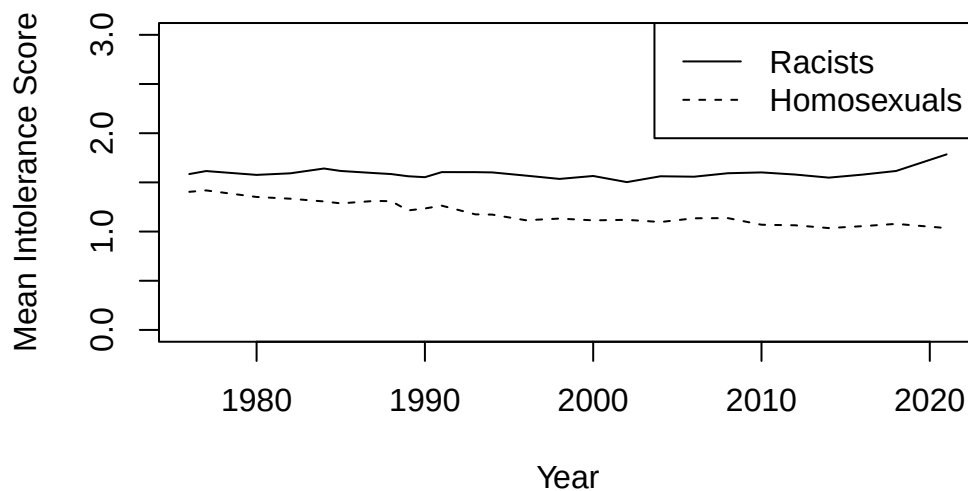
# Create plot
plot(figure2$year, figure2$racist_mean,
     type = "l",
     ylim = c(0, 3),
     main = "Intolerance Towards Racists and Homosexuals, 1976-2024",
     xlab = "Year",
     ylab = "Mean Intolerance Score")

# Add homosexual intolerance line
lines(figure2$year, figure2$homosexual_mean,
      lty = 2)

# Add legend
legend("topright",
      legend = c("Racists", "Homosexuals"),
      lty = c(1, 2))

```

Intolerance Towards Racists and Homosexuals, 1976–2024



Repliate Figure 5

```

# Figure 5: A Dichotomous Measure of Tolerance and Intolerance, 1976-2024
# If a respondent gives at least one intolerant response among the 15 items, they are classified as intolerant

# Create dichotomous measure

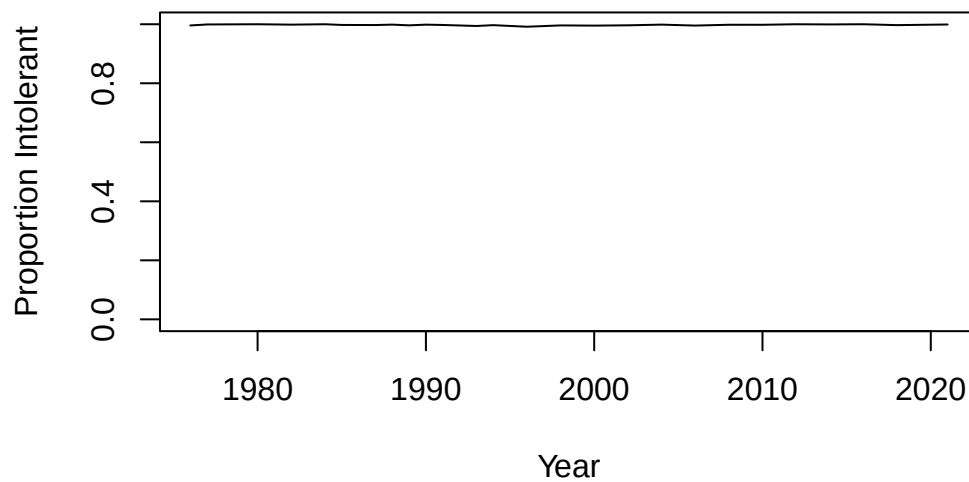
gss <- gss |>
  mutate(intolerant = case_when(
    intolerance_score == 0 ~ 0,
    intolerance_score > 0 ~ 1,
    # If there is a missing value for intolerance_score, keep it missing
    TRUE ~ NA
  ))

# Calculate proportion intolerant by year
figure5 <- gss |>
  group_by(year) |>
  summarize(
    proportion_intolerant = mean(intolerant, na.rm = TRUE)
  ) |> # Remove years with no data
  filter(!is.na(proportion_intolerant))

# Create plot
plot(figure5$year, figure5$proportion_intolerant,
     type = "l",
     ylim = c(0, 1),
     main = "A Dichotomous Measure of Tolerance and Intolerance, 1976-2024",
     xlab = "Year",
     ylab = "Proportion Intolerant")

```

A Dichotomous Measure of Tolerance and Intolerance, 1976-



Original Work

```
# HEALTH

gss <- gss |>
  mutate(health = case_when(
    health == 1 ~ 4,
    health == 2 ~ 3,
    health == 3 ~ 2,
    health == 4 ~ 1,
    # The rest is missing
    TRUE ~ NA_real_
  ))

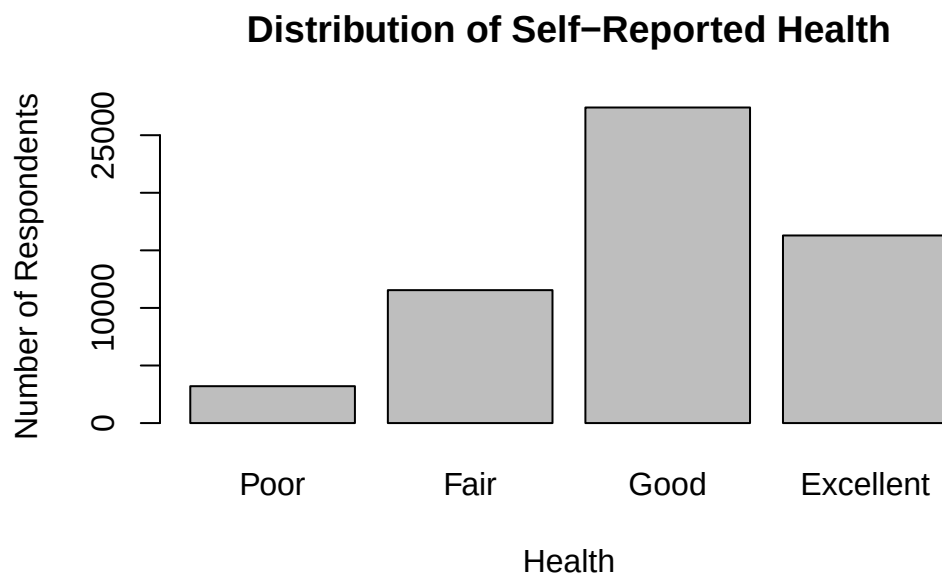
# Central tendency and spread
mean(gss$health, na.rm = TRUE)
```

```
[1] 2.971599
```

```
sd(gss$health, na.rm = TRUE)
```

```
[1] 0.8335249
```

```
# Distribution of Health Responses
barplot(table(gss$health),
        main = "Distribution of Self-Reported Health",
        xlab = "Health",
        ylab = "Number of Respondents",
        names.arg = c("Poor", "Fair", "Good", "Excellent"))
```



```
# HAPPINESS

gss <- gss |>
  mutate(happy = case_when(
    happy == 1 ~ 3,
    happy == 2 ~ 2,
    happy == 3 ~ 1,
    # The rest is missing
    TRUE ~ NA_real_
  ))
```

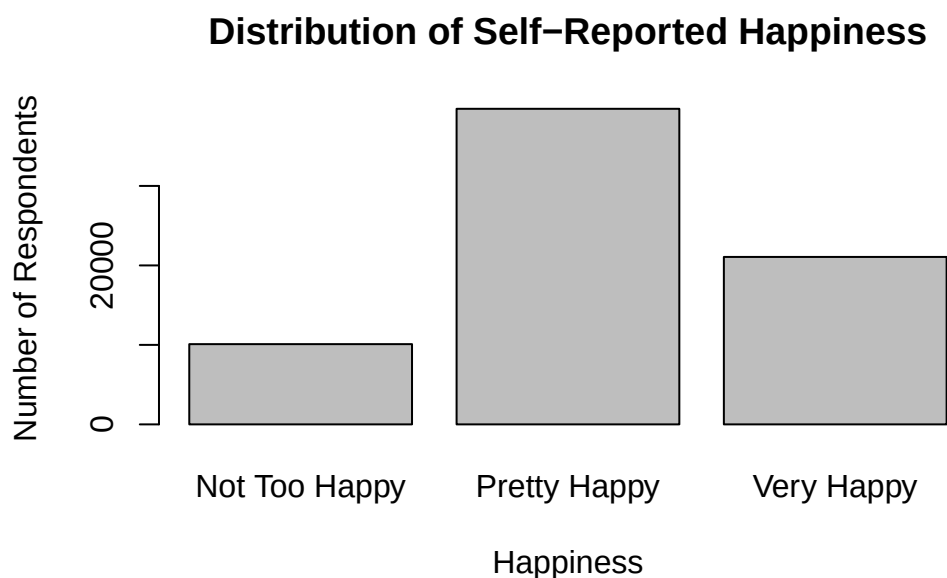
```
# Central tendency and spread  
mean(gss$happy, na.rm = TRUE)
```

```
[1] 2.154849
```

```
sd(gss$happy, na.rm = TRUE)
```

```
[1] 0.6448012
```

```
# Distribution of Happiness Responses  
barplot(table(gss$happy),  
        main = "Distribution of Self-Reported Happiness",  
        xlab = "Happiness",  
        ylab = "Number of Respondents",  
        names.arg = c("Not Too Happy", "Pretty Happy", "Very Happy"))
```



Where did you encounter problems with this replication? Was there anything that was particularly tricky?

I had a difficult time identifying which variables to use in the code book to replicate the figures, as there were over 6,000 columns.

How would you interpret (in other words, explain to a general audience) what Mondak and Sanders found in figures 1, 2, and 5?

Has anything changed since 1998 that would alter the substantive takeaway points in their findings?